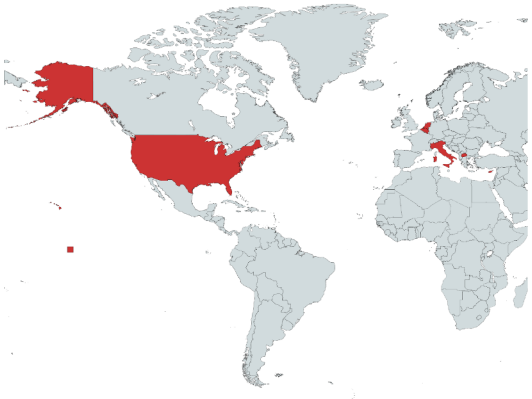
AI, Automation, 4th Industrial Revolution, Jobs

Darko Bozhinoski,
Ph.D. in Computer Science

About Me







Lived and worked in
6 countries

Started Ph.D.

Nov 2013

Ph.D. in Computer Science

Gran Sasso Science Institute, L'Aquila, Italy

Dec 2017

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IRIDIA, Artificial Intelligence, ULB, Brussels, Belgium

Jan 2018

Researcher

Cognitive Robotics, TU Delft, Delft, Netherlands

Jan 2019

Senior Researcher

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Oct 2021

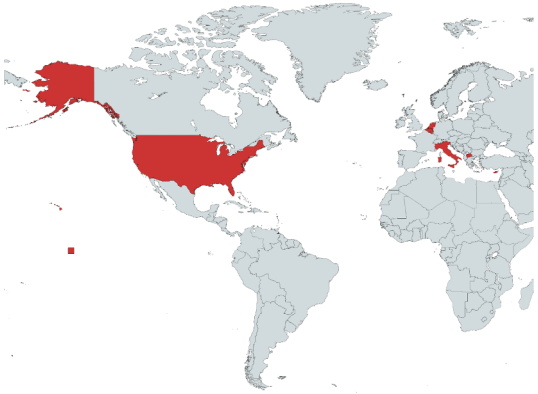
Visiting Professor

InnoPolis, Russia

March 2022







Lived and worked in
6 countries

Research Focus

Autonomous systems

Generative AI

Self-adaptive systems

Formal verification

Single and Multi-robot systems

Robot Swarms

Academic Partners







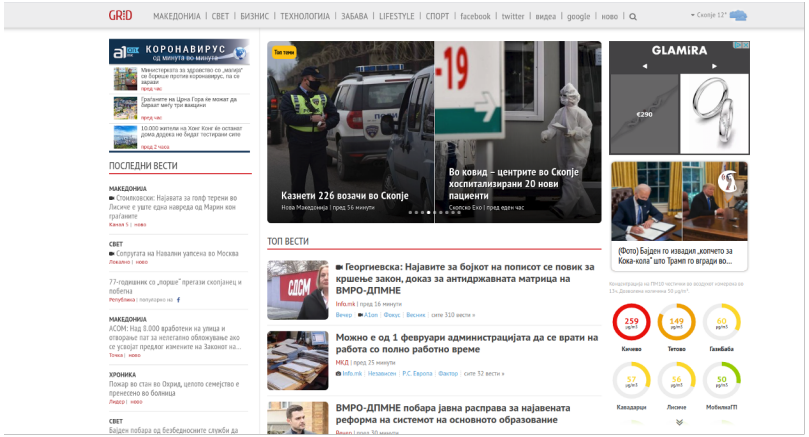






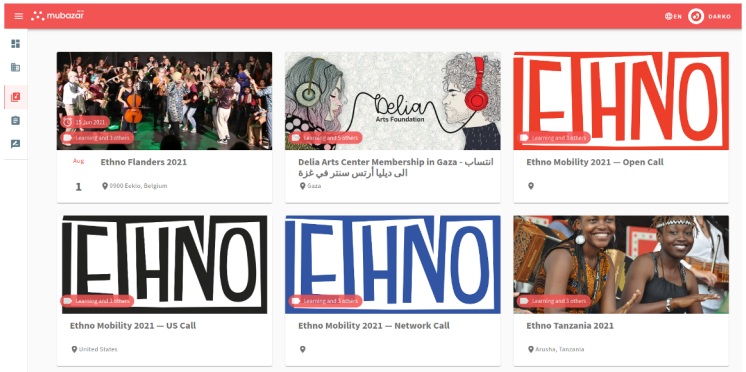


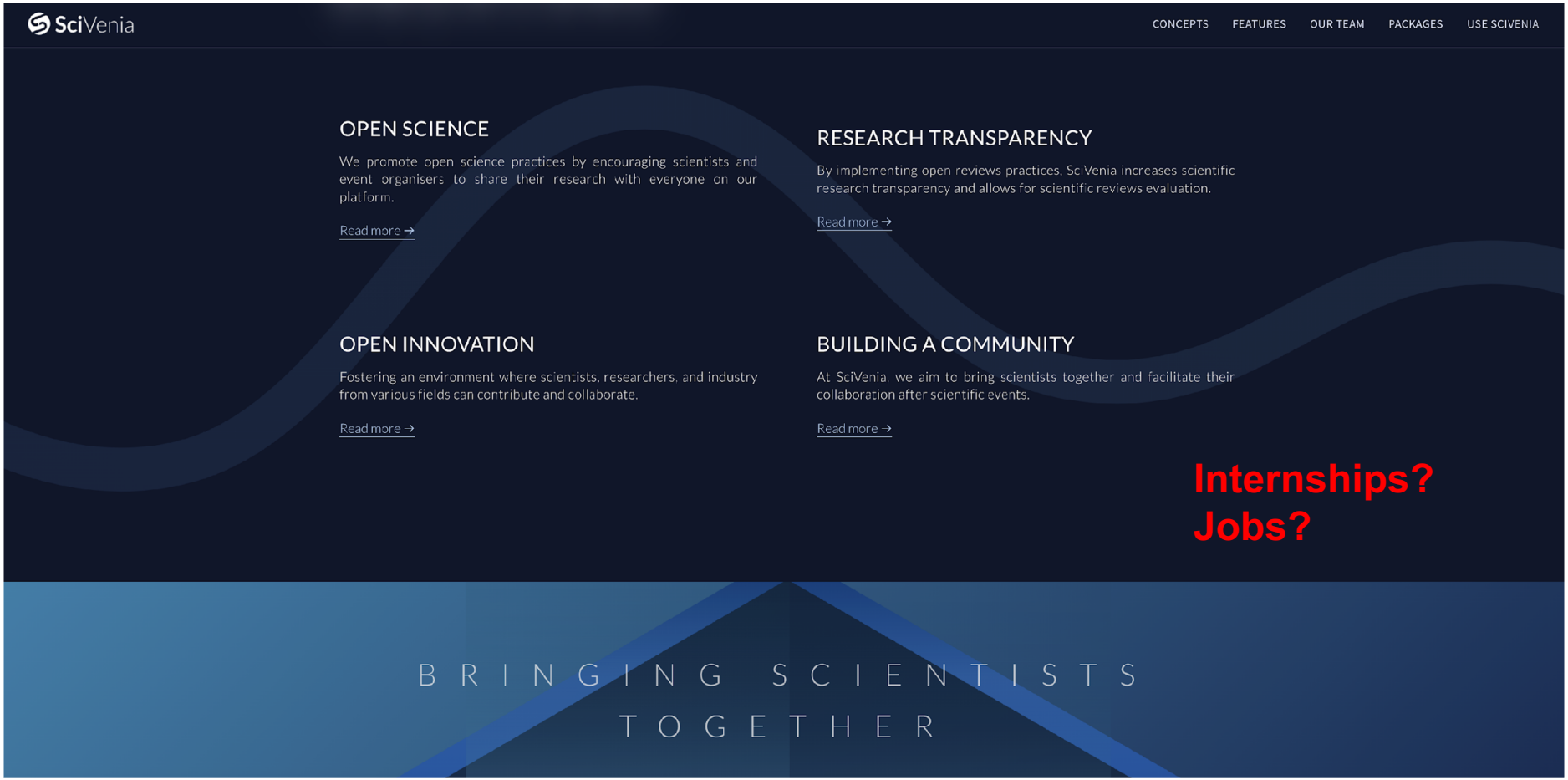
Industrial Projects

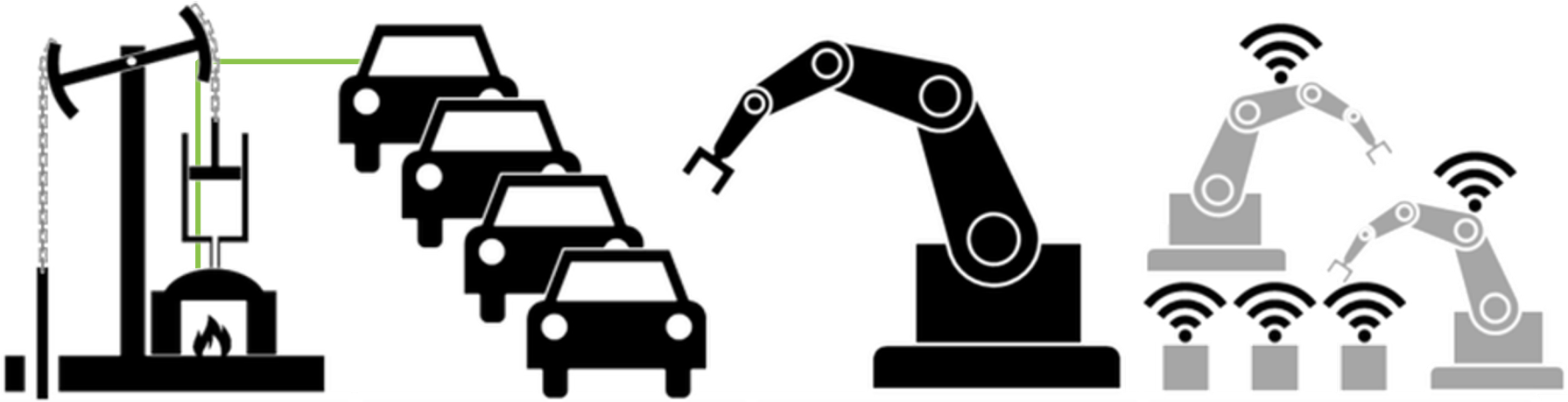




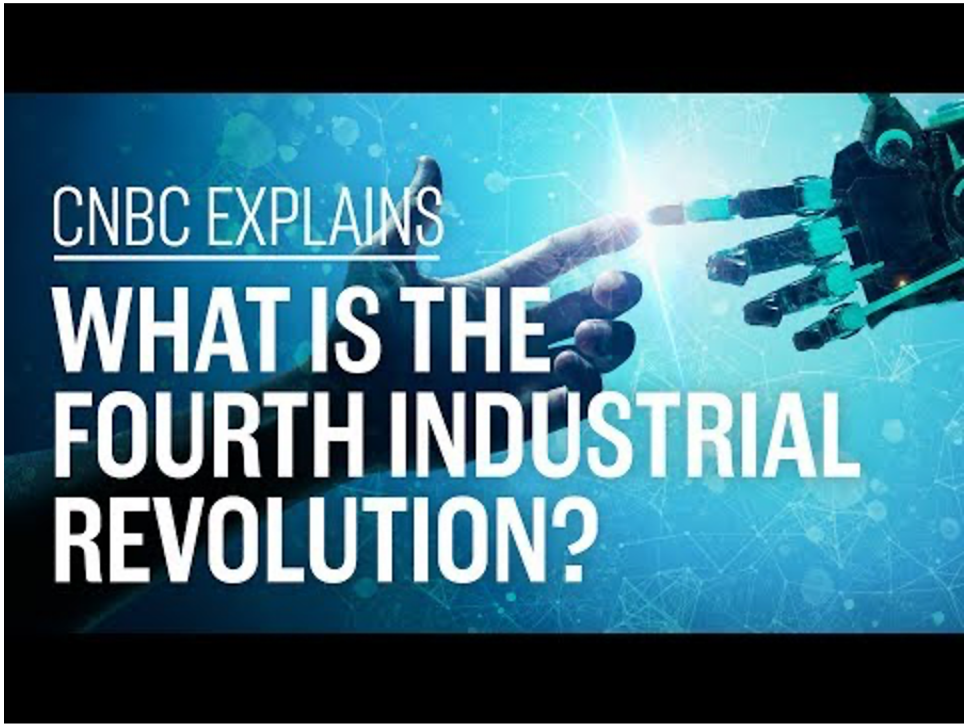
Collaboration with startups;




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BRINGING SCIENTISTS TOGETHER



1st	2nd	3rd	4th
Mechanization, water power, steam power	Mass production, assembly line, electricity	Computer and automation	Cyber Physical Systems


CNBC EXPLAINS
WHAT IS THE FOURTH INDUSTRIAL REVOLUTION?

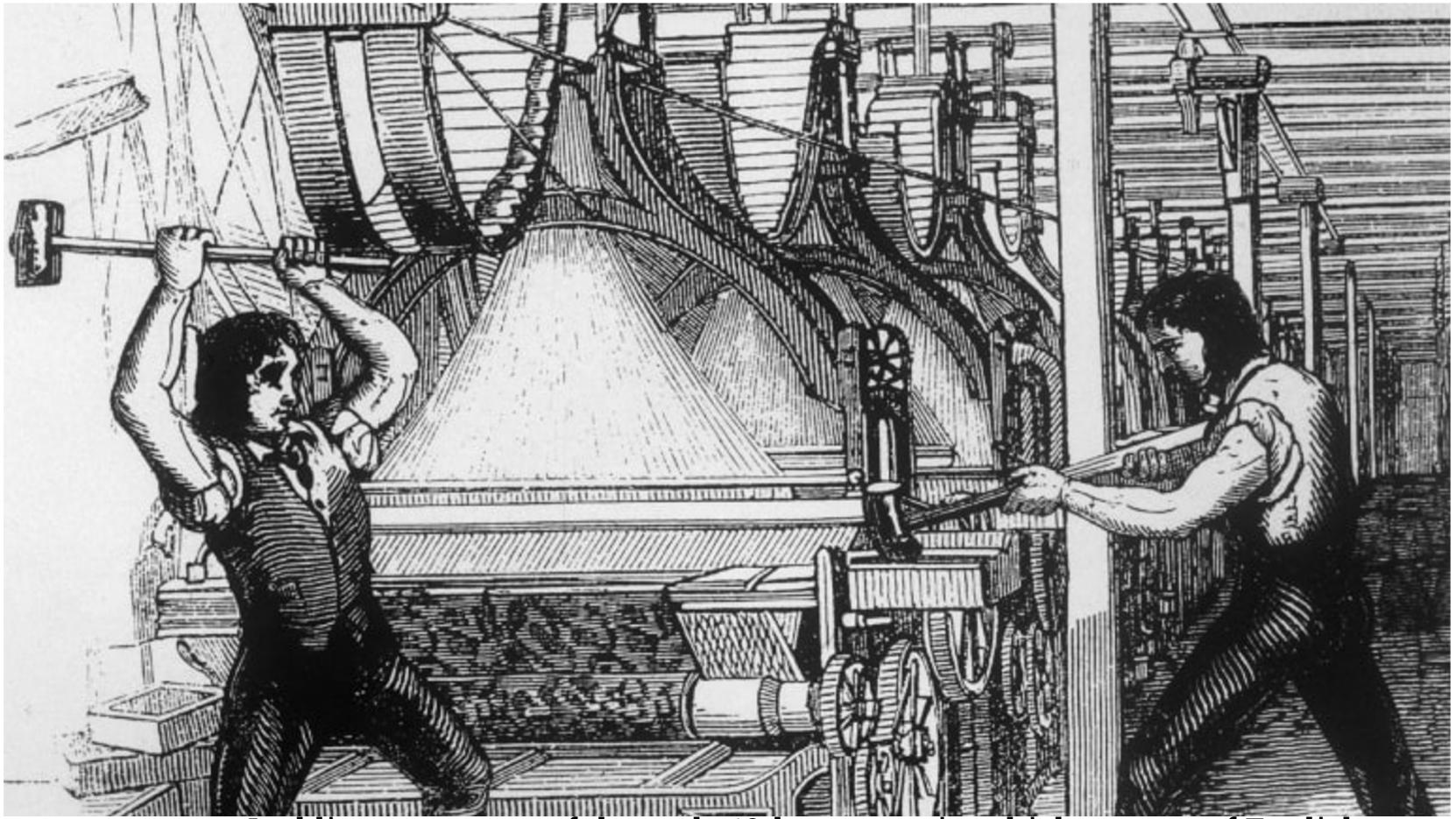
Consequences of the 4th Industrial Revolution

- Create bigger inequality
- Innovators, investors and shareholders benefit the most from the innovation
- AI will eliminate some jobs
- Create demand for new skills that many workers don’t have
- Privacy concerns

Journalists and expert commentators **tend to overstate** the extent of machine substitution for human labor and ignore the strong complementarities between automation and labor that increase productivity, raise earnings, and augment demand for labor.

Why Are There Still So Many Jobs? The History and Future of Workplace Automation
American Economic Association

Have we been here before?



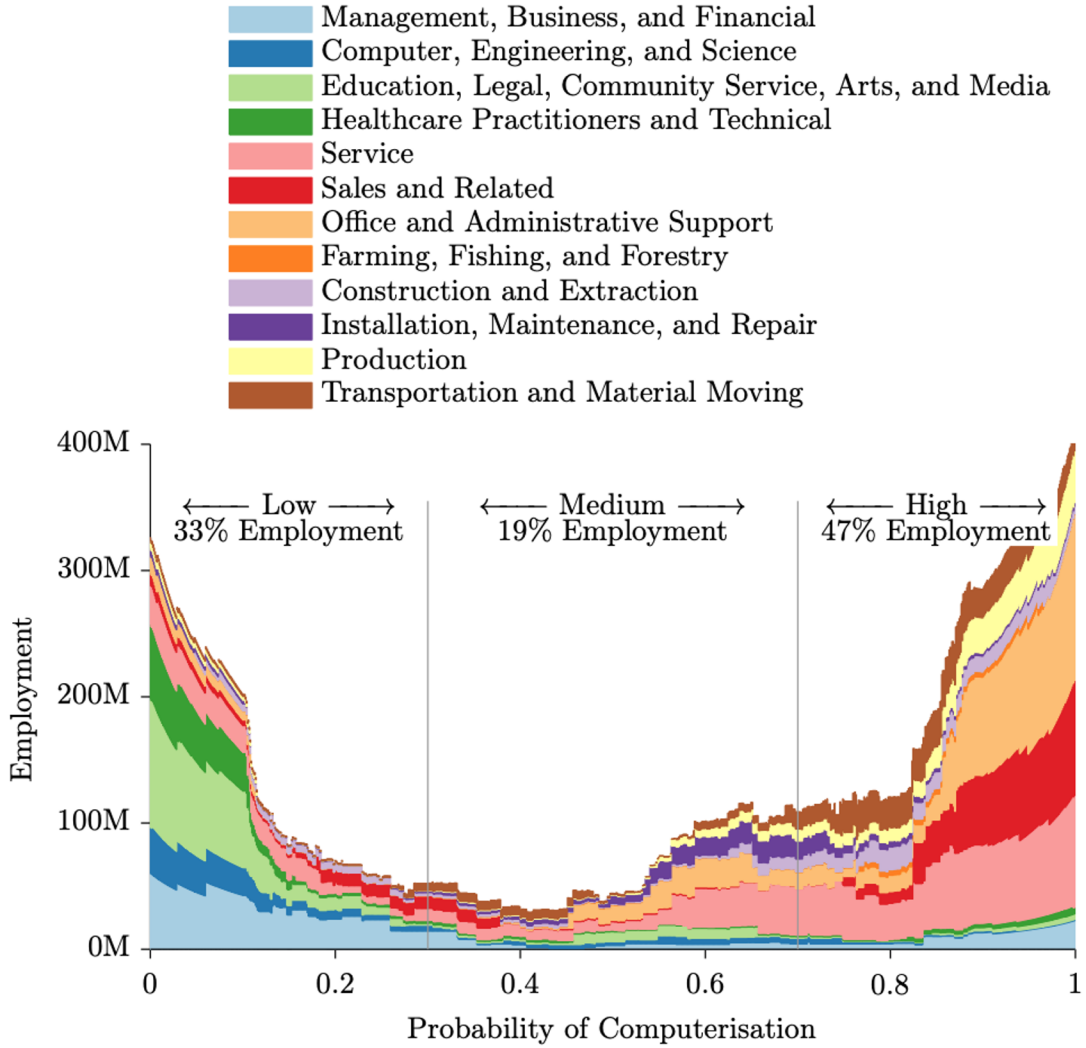
Luddite movement of the early 19th century, in which a group of English textile artisans protested the automation of textile production by seeking to destroy some of the machines.

In the 1960s that president Lyndon B Johnson created a special “Blue-Ribbon National Commission on Technology, Automation, and Economic Progress” to examine the consequences for jobs and living standards. The national commission eventually concluded that mass unemployment was not a risk.

“The basic fact is that technology eliminates jobs, not work”

The past two centuries of Automation and technological progress have not made human labor obsolete.

Should we worry?



47 percent of U.S. jobs are at risk from automation

THE FUTURE OF EMPLOYMENT: HOW SUSCEPTIBLE ARE JOBS TO COMPUTERISATION?

Conceptualizing automation

Jobs are made up of many tasks and that while automation and computerization can substitute for some of them, understanding the interaction between technology and employment requires thinking about about the range of tasks involved in jobs, and how human labor can often complement new technology (is it not a simple substitution.)

The task categorisation of Autor, et al. (2003) distinguishes between workplace tasks using a two-by-two matrix, with routine versus non-routine tasks on one axis, and manual versus cognitive tasks on the other.

Conceptualizing automation

Jobs are made up of many tasks and that while automation and computerization can substitute for some of them, understanding the interaction between technology and employment requires

Smarter together: Why artificial intelligence needs human-centered design

Conceptualizing automation

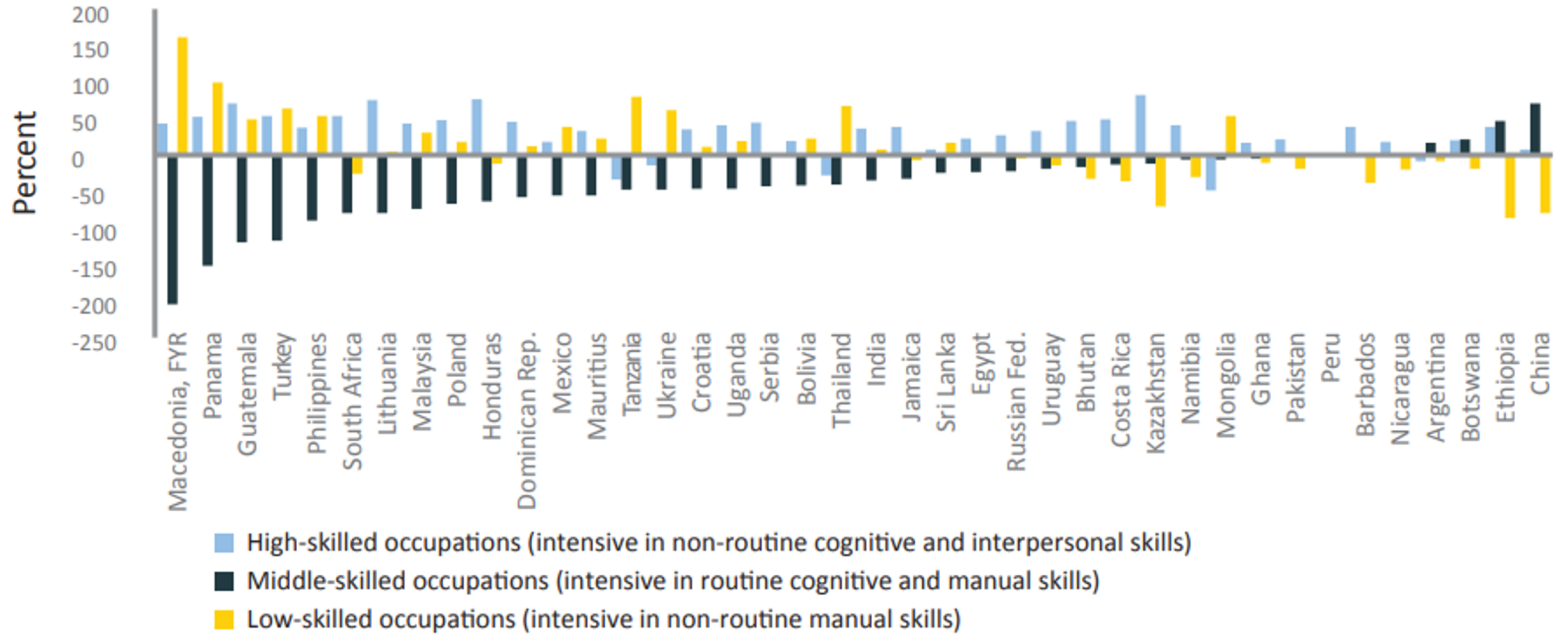
Automation has historically increased workers' productive and enabled humans to transition out of physically difficult, mundane, or menial labor, and in so doing, raised the standard of living.

Conceptualizing automation

Automation has historically increased workers' productive and enabled humans to transition out of physically difficult, mundane, or menial labor, and in so doing, raised the standard of living.

In the United States between 1900 and 2000, farming went from being the main employer in the economy, with 41 percent of all jobs, to employing only 2 percent of workers, according to data from the U.S. Department of Agriculture.

Figure 1. Annual Average Change in Employment Share, circa 1995–2012



Source: World Bank 2016.

FIGURE 22

Top 20 job roles in increasing and decreasing demand across industries

↗ Increasing demand

1	Data Analysts and Scientists
2	AI and Machine Learning Specialists
3	Big Data Specialists
4	Digital Marketing and Strategy Specialists
5	Process Automation Specialists
6	Business Development Professionals
7	Digital Transformation Specialists
8	Information Security Analysts
9	Software and Applications Developers
10	Internet of Things Specialists
11	Project Managers
12	Business Services and Administration Managers
13	Database and Network Professionals
14	Robotics Engineers
15	Strategic Advisors
16	Management and Organization Analysts
17	FinTech Engineers
18	Mechanics and Machinery Repairers
19	Organizational Development Specialists
20	Risk Management Specialists

↘ Decreasing demand

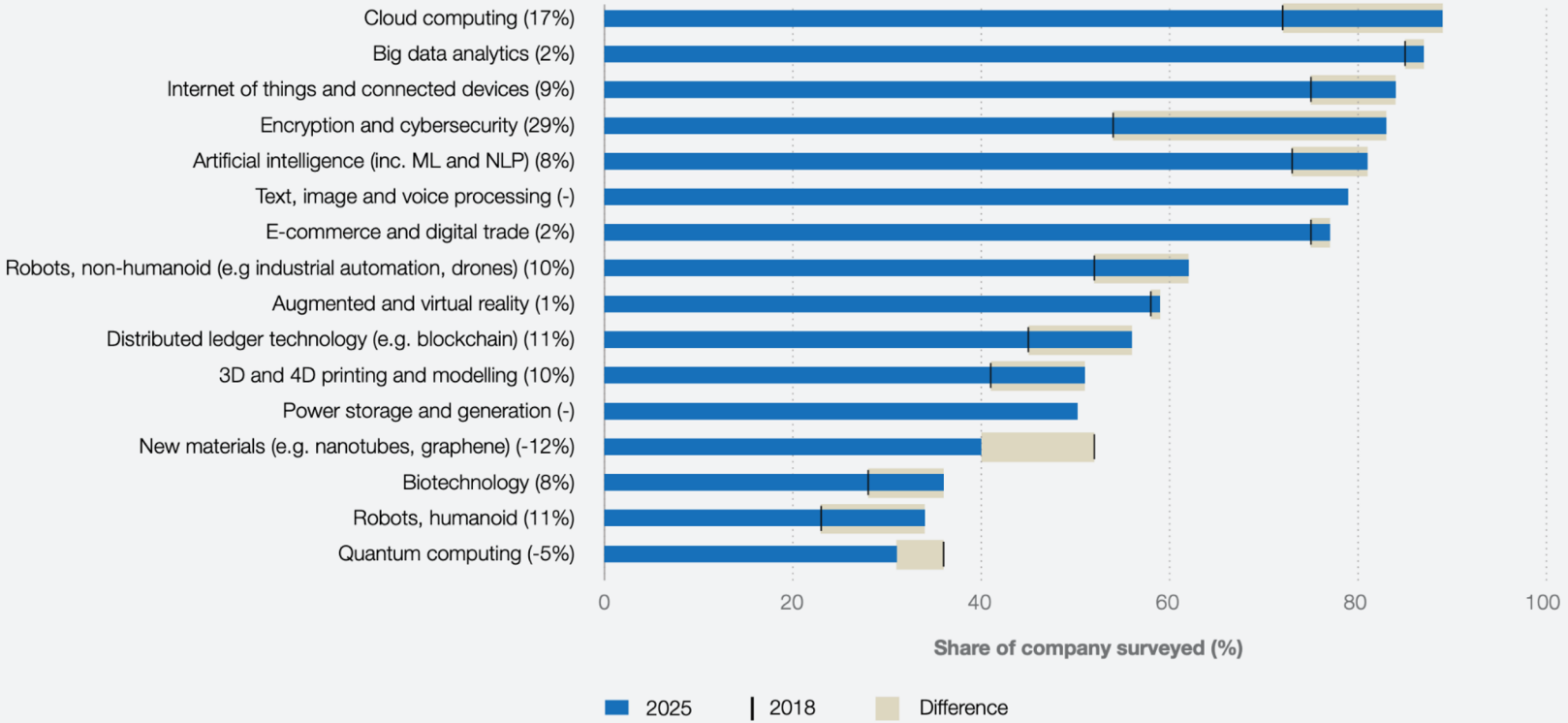
1	Data Entry Clerks
2	Administrative and Executive Secretaries
3	Accounting, Bookkeeping and Payroll Clerks
4	Accountants and Auditors
5	Assembly and Factory Workers
6	Business Services and Administration Managers
7	Client Information and Customer Service Workers
8	General and Operations Managers
9	Mechanics and Machinery Repairers
10	Material-Recording and Stock-Keeping Clerks
11	Financial Analysts
12	Postal Service Clerks
13	Sales Rep., Wholesale and Manuf., Tech. and Sci.Products
14	Relationship Managers
15	Bank Tellers and Related Clerks
16	Door-To-Door Sales, News and Street Vendors
17	Electronics and Telecoms Installers and Repairers
18	Human Resources Specialists
19	Training and Development Specialists
20	Construction Laborers

Source

Future of Jobs Survey 2020, World Economic Forum.

FIGURE 18

Technologies likely to be adopted by 2025 (by share of companies surveyed)



Source

Who is affected?

In the recent past, those hardest hit were men with low levels of education who dominated manufacturing and other blue-collar jobs, and women with intermediate levels of education who dominated clerical and administrative positions.

https://willrobotstakemyjob.com/

TABLE I. O*NET variables that serve as indicators of bottlenecks to computerisation.

Computerisation bottleneck	O*NET Variable	O*NET Description
Perception and Manipulation	Finger Dexterity	The ability to make precisely coordinated movements of the fingers of one or both hands to grasp, manipulate, or assemble very small objects.
	Manual Dexterity	The ability to quickly move your hand, your hand together with your arm, or your two hands to grasp, manipulate, or assemble objects.
	Cramped Work Space, Awkward Positions	How often does this job require working in cramped work spaces that requires getting into awkward positions?
Creative Intelligence	Originality	The ability to come up with unusual or clever ideas about a given topic or situation, or to develop creative ways to solve a problem.
	Fine Arts	Knowledge of theory and techniques required to compose, produce, and perform works of music, dance, visual arts, drama, and sculpture.
Social Intelligence	Social Perceptiveness	Being aware of others' reactions and understanding why they react as they do.
	Negotiation	Bringing others together and trying to reconcile differences.
	Persuasion	Persuading others to change their minds or behavior.
	Assisting and Caring for Others	Providing personal assistance, medical attention, emotional support, or other personal care to others such as coworkers, customers, or patients.

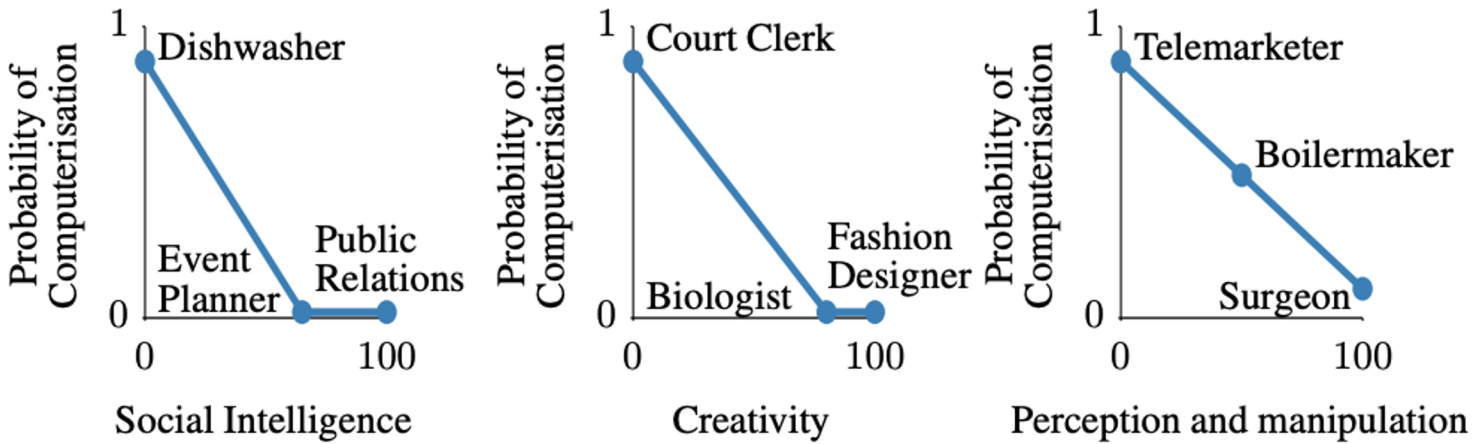


FIGURE I. A sketch of how the probability of computerisation might vary as a function of bottleneck variables.

What can we say about the current technologies?

While in the past, technology was increasingly able to perform routine manual and cognitive tasks, in the current digital and computing revolution, machines can also perform nonroutine tasks that had been reserved to humans.

The replacement of labor by machines takes time and depends on circumstances specific to a given context. Technological innovations tend to occur in developed countries, and their adoption in developing countries usually occurs with a time lag.

Man Automated His Job One Year Ago And Is Still Earning $90,000

Dominic Smithers

Published 13:07, 13 January 2022 at GMT
| Last updated 13:07, 13 January 2022 at GMT



The pandemic forced millions of people to change their working habits, moving from the office to the bedroom and only seeing colleagues over Zoom.

But while many of us struggled to adapt to the 'new normal', one man managed to make it work to his advantage.

An IT specialist recently revealed that he was able to make his $90,000 (£65,603) a year job fully automated.

How to mitigate the negative effects of technological change?

How to mitigate the negative effects of technological change?

How to maximize the potential social gains from technological change?

Government perspective

- Policies that will facilitate labor market flexibility and mobility
- Policies that introduce and strengthen safety nets and social protection
- Improved education and training process.
  - Complementing fundamental education with active labor market policies,
  - workforce training
  - lifelong learning opportunities that engage workers to participate in the changing labor market

Company Perspective

- Effective transformation of the production process
- Catching up with latest technologies
- Strategies for retraining the company stuff

Amazon will retrain one-third of its U.S. employees to get ahead of tech changes

$700 million initiative that will cover 100,000 workers by 2025.

Beth Galetti, Amazon's senior vice president for HR. "We think it's important to invest in our employees, and to help them gain new skills and create more professional options for themselves."

Business

Amazon will retrain one-third of its U.S. employees to get ahead of tech changes

Individual Perspective



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2. The future of employment: How susceptible are jobs to computerisation?
3. The Future of Work: Race with—not against—the Machine